Energy meter

EM-001
HOW IT WORKS
VOLTS TIMES AMPS, ADDED UP

How do you read the units on this meter?

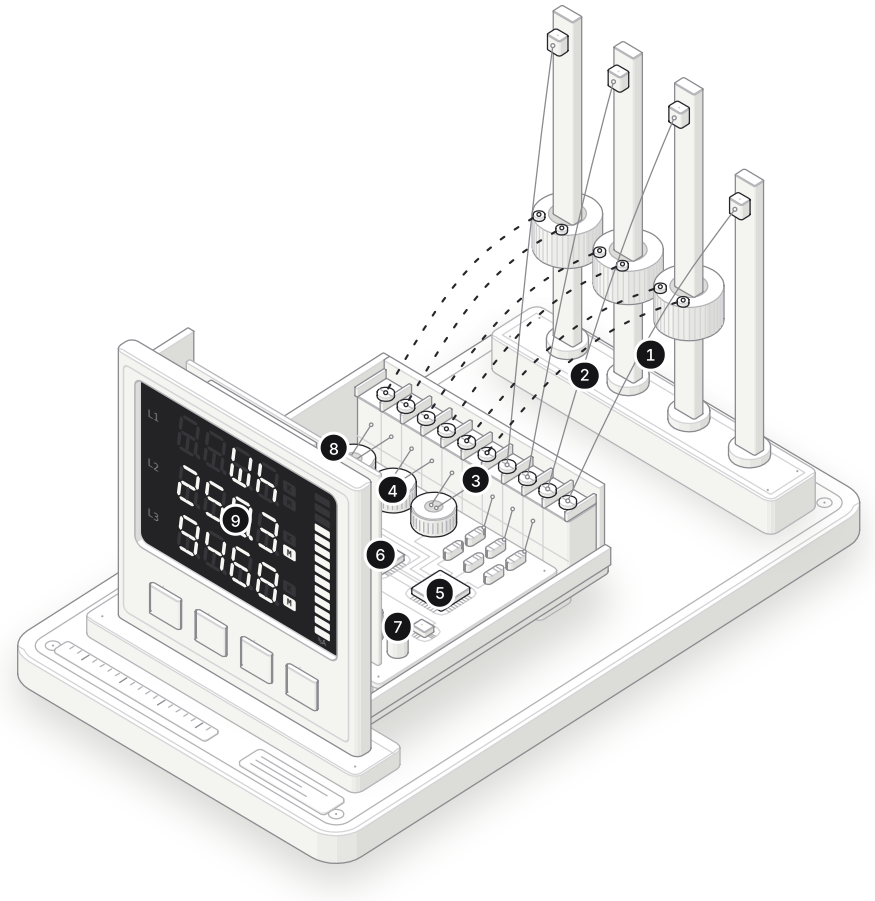
CTs and taps bring each phase's current and voltage in; the metering chip multiplies them into power and the MCU adds it up into Wh. Rows 2 and 3 of the window are one 8-digit number, and the lit K or M lamp gives kWh or MWh.

258.39468 MWh
ON THE WINDOW

2,58,394.68 units
THE SAME IN KWH

400 : 5
CT RATIO

186.3 kW
AT 300 A PER PHASE



01 **Bus bars and taps**
L1 L2 L3 carry the load; taps send 230 V.

02 **External CTs 400 : 5**
Step bar current down to 0–5 A.

03 **Terminal block**
I1–I3′ take the CT loops, V1–VN the voltages.

04 **Internal CTs, divider**
Scale current and voltage down for the chip.

05 **Metering IC**
Works out P = V × I × cos φ, three phases.

06 **MCU, crystal, EEPROM**
Add power over time into Wh and keep it.

07 **Power supply**
Transformer and capacitors run the meter.

08 **Display board**
Carries the total to the three LED rows.

09 **Window and keys**
Wh row, 8-digit energy, K/M lamps; keys page.